Ganit

data speaks

HDFC Life – GenAI
Knowledge Bot

Case Study

Transforming Insurance Operations with a Multilingual GenAI-Powered Knowledge Assistant

May 2026

300–400 SOPs & FAQs Indexed	2 Live Operational Datamarts	English + Hindi Multilingual	AWS Cloud Native Architecture	Citation-Backed Responses	SSO-Authenticated Chat UI

The Challenge

HDFC Life's operations teams managed day-to-day queries by manually scanning 300–400 SOP/FAQ documents across PDFs, Word, and Excel files — many containing embedded flowcharts, images, and tabular data. With no centralized search or conversational interface, teams depended on individual expertise ('tribal knowledge'), leading to inconsistent answers, resolution delays of hours, and a complete absence of multilingual support for English and Hindi-speaking staff. Two operational Datamarts held live business data, but querying them required technical skills unavailable to frontline operations staff.

The Solution

Ganit designed and deployed a GenAI-powered Knowledge Bot on AWS Cloud, combining Retrieval-Augmented Generation (RAG) over indexed SOP/FAQ documents with live structured queries against Datamarts. The bot uses Claude Sonnet 3.5 / OpenAI models via API Gateway and AWS Lambda, stores conversation history in DynamoDB, and authenticates users via Cognito with Microsoft SSO. The React-based frontend delivers streaming, citation-backed, multilingual responses directly within HDFC Life's existing operations ecosystem.

Results Achieved

70%+

Reduction in avg. query resolution time

300–400

SOP/FAQ documents fully indexed & searchable

2

Live Datamarts integrated with natural language access

EN+HI

Bilingual responses for all operations staff

100%

Responses carry mandatory source citations

Zero

Tribal knowledge dependency for standard queries

Modernizing Operations Knowledge Access for HDFC Life with GenAI — Enabling Real-Time Query Resolution Across SOPs, FAQs & Live Datamarts

Problem Statement

- 300–400 SOPs & FAQs in PDFs, Word & Excel — manual search only
- No centralized interface — tribal knowledge dependency
- No English/Hindi multilingual conversational support
- 2 Datamarts with zero natural-language query interface
- No chat history or per-user session continuity

Solution Overview

HDFC Life needed a unified GenAI bot ingesting 300–400 SOP/FAQ documents and 2 live Datamarts to drive instant, citation-backed, multilingual query resolution for frontline operations teams.

SOP / FAQ / Datamart Sources

- AWS WAF + CloudFront + React UI
- API Gateway + Lambda (Fast API)
- Claude Sonnet 3.5 / OpenAI + Cohere
- OpenSearch / pgvector (RAG Engine)
- DynamoDB — Chat + Knowledge Store
- Cognito SSO + Secrets Manager

Application UI

- ✓ Streaming chat with thinking indicator
- ✓ Source citations (SOP name + version)
- ✓ English / Hindi language toggle
- ✓ Persistent conversation history
- ✓ Multimodal — text, images, flowcharts
- ✓ Video links for step-by-step guidance
- ✓ Regenerate response option

AWS Stack: WAF · CloudFront · API Gateway · Lambda · OpenSearch · DynamoDB · S3 · Cognito · ECS/EKS · Secrets Manager · CloudWatch

Business Value Achieved

- ✓ 70%+ faster query resolution vs manual SOP scanning
- ✓ Centralized knowledge eliminating SME escalation
- ✓ Bilingual EN/HI coverage for all operations staff
- ✓ Citation-backed answers ensuring IRDAI compliance
- ✓ Natural-language Datamart access for TAT/SLA queries

Ganit unified HDFC Life's knowledge assets on a GenAI platform, accelerating operations query resolution and driving confident decisions across all business functions

Outcome

Ganit delivered a cloud-native GenAI Knowledge Bot for HDFC Life on AWS, ingesting 300–400 SOP/FAQ documents (PDFs, Word, Excel with embedded images and flowcharts) and 2 live operational Datamarts through a RAG-powered orchestration layer using Claude Sonnet 3.5/OpenAI models — surfaced through a React-based multilingual chat UI integrated with HDFC Life's Microsoft SSO for self-service query resolution across all operations functions.

Operational Efficiency

70%+

avg. resolution time cut

Eliminated manual SOP scanning and tribal knowledge dependency by automating document retrieval via RAG pipelines and OpenSearch vector indexing — reducing frontline operations query resolution from hours to seconds with standardized, citation-backed answers.

Cost Optimization

~60%

SME escalation reduction

Unified SOPs, FAQs, and live Datamart access on a single serverless AWS platform (Lambda + ECS), significantly reducing unplanned SME escalations and operational support overhead — with auto-scaling handling peak query volumes without additional headcount.

Scalability & Reliability

300–400

docs indexed with versioning

Built a vector-indexed knowledge base (OpenSearch/pgvector) covering 300–400 versioned SOP/FAQ documents with multimodal content — text, images, flowcharts, and video metadata — handling concurrent multilingual (EN/HI) queries with sub-3-second response times.

Business Growth

100%

IRDAI-cited responses

Enabled real-time, grounded responses from live Datamarts in natural language — giving operations teams instant access to underwriting TAT norms (IRDAI: 15 days), PASA process guidance, policy issuance workflows, and Aadhaar KYC steps with full audit trails.

Fragmented SOPs, siloed Datamarts, and zero multilingual interface blocked unified knowledge access for HDFC Life's operations teams

Problem Statement

HDFC Life's operations teams manage thousands of daily queries across underwriting, policy issuance, KYC, claims, and customer service. Resolution required manually navigating 300–400 PDF/Word/Excel SOPs and FAQs — many containing embedded process flowcharts, scanned images, and multi-step decision trees. Two operational Datamarts held live business data on policies, TAT metrics, and case statuses, but querying them required SQL skills unavailable to frontline staff. With no centralized interface, teams relied on individual expertise, causing resolution delays of 2–8 hours, inconsistent IRDAI-compliant answers, and an inability to serve Hindi-speaking operations staff effectively.

- › 300–400 SOPs/FAQs in PDFs, Word & Excel with embedded flowcharts, images, and scanned tables
- › 2 operational Datamarts (underwriting + policy) accessible only via SQL — not available to ops staff
- › No centralized retrieval engine — teams manually cross-reference 5–10 docs per query
- › Zero multilingual support — Hindi-speaking ops staff translate SOPs manually with errors
- › No chat history or personalized sessions — every query starts from scratch
- › Tribal knowledge concentration: 3–4 SMEs handling 70%+ of escalated queries daily

Challenge 1	Fragmented SOP & FAQ documents with no automated pipeline — ops staff manually scan PDFs and Word files containing multi-page IRDAI-regulated process flows, underwriting SOPs (e.g. Group UW v1.1), and KYC checklists
Challenge 2	Manual document scanning causing 2–8 hour query resolution delays — IRDAI mandates 15-day TAT for underwriting, but internal targets are 2 days; delayed answers directly impact compliance SLA adherence
Challenge 3	Datamart data (TAT metrics, policy statuses, PASA case flags) stranded in operational systems with no natural-language interface — frontline staff cannot self-serve structured business data
Challenge 4	No multilingual (EN/HI) or multimodal support — ops staff cannot get image-based process explanations or access training video metadata; complex workflows like PASA processing are especially hard to navigate verbally

Ganit built a multi-source GenAI Knowledge Bot on AWS — RAG-powered ingestion from SOPs, FAQs & Datamarts, served through a multilingual streaming chat UI

Solution Overview

Robust & Scalable Architecture

› SOP/FAQ documents (PDF, Word, Excel) chunked & embedded into OpenSearch/pgvector via ECS batch jobs with incremental delta refresh on every document version update.

› Datamart REST APIs polled via Lambda on schedule with structured query routing — results merged with RAG context before LLM generation.

› Claude Sonnet 3.5 / OpenAI GPT-4 as LLM backbone with Cohere reranking; citation enforcement via system-level prompt engineering on every response turn.

› DynamoDB stores conversation history, knowledge base metadata, and WebSocket session state — enabling per-user personalization and query continuity across sessions.

Application UI — Chat Interface

› React (TypeScript) frontend delivered via CloudFront CDN, authenticated through AWS Cognito with Microsoft SSO — no separate login for HDFC Life ops staff.

› Streaming responses via API Gateway WebSockets + Fast API Lambda; source citation panel shows SOP name, version, and page reference.

› Language toggle (EN/HI) with multimodal responses: text summaries + embedded process flowchart images + related training video links from S3 metadata.

AWS Cloud Architecture

FRONTEND · AWS WAF · CloudFront · React Application

KNOWLEDGE BASE STORAGE · S3 Documents · Chat History DB

STREAMING LAYER · API Gateway WebSockets · Lambda (Fast API)

AI MODELS · Claude Sonnet 3.5 / OpenAI · Cohere Reranker

RAG ENGINE · OpenSearch · pgvector · ECS Embedding Jobs

APP DATABASES · DynamoDB (Conversations · KB Meta · WebSocket)

AUTH & SECURITY · Cognito (SSO) · Secrets Manager · CloudWatch

3 Query Workflows | 50+ Operational KPIs | 300–400 Docs | EN+HI | 3 Source Systems | Sub-3s Response

Query Resolution Workflow — How the GenAI Bot Answers a Live Operations Query (Example: 'What is the TAT process for underwriting?')

Business Problem

› Operations teams rely on multiple SOPs, FAQs, and internal Datamarts to resolve queries — fragmented access creates longer turnaround times and inconsistent responses.

› Manual search across documents and databases creates dependency on individual expertise, inconsistent responses, and operational inefficiencies that directly impact IRDAI SLA compliance.

› Lack of a unified multilingual support interface limits scalability, slows decision-making, and impacts service quality across English and Hindi-speaking operations teams.

Impact Generated

› Centralized GenAI assistant delivers instant, citation-backed responses from enterprise knowledge and live Datamart data — e.g. TAT norms (IRDAI: 15 days, Internal target: 2 days) surfaced in <3 seconds.

› Reduced query resolution time from 2–8 hours to under 3 seconds through automated RAG retrieval, real-time LLM generation, and standardized IRDAI-compliant guidance.

› Improved productivity through multilingual support, persistent chat history per user, and secure SSO access — eliminating the need for SME escalation for standard process queries.

Live Workflow — TAT Process Query

Step 1

User Query Submitted

Operations staff types 'What is the TAT process?' into the chat UI. The React frontend sends the query via API Gateway WebSocket to the Lambda backend with session context (user ID, conversation history, language=EN).

Step 2

AI Reasoning & RAG Retrieval

Orchestration layer checks Datamart APIs for live TAT metrics, then queries OpenSearch vector index for relevant SOPs. 'SOP Group UW v1.1 261225' is retrieved as the primary source. Cohere reranks top-5 chunks. LLM (Claude Sonnet 3.5) generates a grounded response.

Step 3

Citation-Backed Final Response

Bot returns: 'TAT (Turn-Around-Time) measures process completion speed. IRDAI mandates a maximum of 15 days for underwriting overall; the internal team targets 2 days.' Source: SOP Group UW v1.1 261225. User can expand source panel, regenerate, or ask a follow-up.

Multimodal Query Resolution — Visual Flowcharts, Process Images & Training Video Links Delivered Directly in Chat (Example: 'Tell me about PASA process')

Business Problem

› Critical process knowledge locked in flowcharts, scanned process diagrams, and training videos — inaccessible during live queries (e.g. PASA — Proposal Deposit & Suspense Management).

› Users manually search across PDFs, SharePoint folders, and video libraries to find the right text, image, or video reference — averaging 15–30 min for complex process queries.

› Traditional text-only support models lack visual guidance for multi-step operational processes like PASA case routing, KYC validation, and policy issuance workflows.

Impact Generated

› First-time query resolution increases significantly — bot delivers the exact SOP section, embedded process flowchart, and a related training video link in a single response, eliminating the need to search multiple sources.

› Improved understanding of multi-step workflows (PASA, OPUS underwriting, Aadhaar KYC) via visual aids — process images extracted from SOP PDFs are served inline within the chat response.

› Strengthened self-service adoption: frontline ops staff resolve PASA cases, KYC exceptions, and TAT queries independently — reducing SME escalations for repetitive process guidance.

📄 Text + Source Citation

Query: 'Tell me about PASA process'

Bot retrieves SOP NB OTC Processing of PASA Cases v1.0 261225 and returns:
• PASA (Proposal Deposit & Suspense Management) handles proposals flagged for PASA or Penny Drop failures
• Workflow identifies cases, reviews bank proof, matches account holder, validates PAN/Aadhaar, then routes to policy issuance or manual QC
• Source: SOP NB OTC Processing of PASA cases v1.0 261225

🖼 Visual Process Flowchart

Bot extracts and renders the embedded PASA process flowchart from the SOP PDF inline in chat:
• Application Login → Analytics API → Find PASA case → Case processed/manual OTC
• Doc QC user checks documents, validates LOV, updates OPUS
• If all LOV responses 'Yes' → automatic policy issuance; else → exits PASA flow
Users see the exact process diagram without opening the SOP document.

🎥 Training Video + Related Query

Query: 'How to change Aadhaar address?'

Bot returns text answer + video link:
• Source: Video Metadata v01 — 'A training video explains how to update the Aadhaar card address'
• Video title: 'Aadhaar_Card__Address_change_01092025_V_01'
• Watching the video guides through required steps for Aadhaar card address change on the UIDAI portal
• Related video suggested: Aadhaar_Card__Address_change_0109_V_01

Knowledge Design & Chatbot Training Framework — How the Bot Is Built to Be Accurate, Compliant, and Continuously Improving

Training Framework

1. Knowledge Ingestion Pipeline

- Document parsing: PDFMiner + Textract for scanned PDFs; python-docx for Word; openpyxl for Excel Datamarts
- Chunking strategy: 512-token overlapping chunks (128-token overlap) preserving section headers as metadata for citation accuracy
- Embedding model: Cohere Embed v3 / text-embedding-3-large — multilingual embeddings supporting EN+HI in same index
- Vector store: OpenSearch Serverless with HNSW algorithm; cosine similarity for retrieval; metadata filters (SOP version, date, department)
- Delta refresh: ECS batch job monitors S3 for new/updated SOP uploads — re-embeds only changed chunks, preserving stable doc IDs
- Datamart connector: Lambda polls REST APIs every 15 min; structured results cached in DynamoDB with TTL for fresh TAT/SLA data

2. Prompt Engineering & Guardrails

- System prompt mandates: (a) answer only from retrieved context, (b) always cite source name + version, (c) flag if answer not found — never hallucinate
- Citation enforcement: LLM instructed to include 'Source: [SOP Name vX.X XXXXXX]' in every factual claim — validated via output parser before streaming to UI
- Multilingual instruction: language detection on query; system prompt appended with 'Respond in Hindi' if detected — same RAG pipeline, language-specific generation
- Fallback chain: if RAG confidence score <0.7, bot responds 'I could not find a definitive answer in the current knowledge base — please consult your team lead or the relevant SOP directly'
- Role-based access: Cognito user groups map to document categories — underwriting staff see UW SOPs; claims staff see claims SOPs only
- Toxicity & PII filters: AWS Comprehend pre-screens inputs; PII (Aadhaar, PAN, mobile) masked in conversation logs for compliance

3. Evaluation & Continuous Improvement

- Human-in-loop review: SMEs rate bot responses weekly (thumbs up/down in UI) — low-rated answers flagged for SOP review or prompt tuning
- Retrieval metrics: precision@3 and NDCG tracked per query category (underwriting, claims, KYC, policy) — target P@3 > 0.85
- Faithfulness evaluation: automated LLM-as-judge checks whether generated answers are grounded in retrieved chunks — hallucination rate target <2%
- SOP version management: document versioning in S3 ensures bot always serves the latest approved SOP; superseded versions archived but traceable
- Query analytics: CloudWatch dashboard tracks top-20 query categories, avg. response time, citation hit rate, and unanswered query rate — fed back to SOP authoring teams
- Monthly retraining trigger: if unanswered query rate >5% for any category, SOP authoring team is notified to create or update the relevant document

Design Principles: Citation-First · Hallucination-Free · Role-Based Access · Version-Controlled SOPs · Multilingual-Native · Human-Reviewed · IRDAI-Compliant Responses